

The Impact of Placement Exams on Refugee Children’s School Enrollment

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Abstract

When enrolling displaced children, schools can face different challenges: absence of school documents, language differences, and prolonged gaps in education. While placement exams can help to assess students’ pre-enrollment knowledge and inform grade assignment, they may, depending on their implementation, also work as an extra barrier to accessing education. This paper studies the effects of introducing mandatory placement exams for Venezuelan refugee children prior to their school enrollment in Northern Brazil. We explore the national administrative enrollment data to estimate a triple difference framework using the policy’s geographic implementation and its targeted age range (8 to 14). We find that the policy has reduced school enrollment of around 1,000 refugee children between 2020 and 2024, with children living in shelters, an especially vulnerable group, likely being more affected. Importantly, using exam registry administrative data, we find that the main obstacle was not families failing to register their children for the exam, but high exam absenteeism rates (30%) caused by long waiting times and difficulties reaching families to inform exam’s date and time.

Keywords: Refugee Children; Access to Education; School Enrollment; Placement Exams; Venezuelan Migration

JEL Codes: I25, I28, J15, O15, F22

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1 Introduction

Between 2010 and 2024, the number of displaced children worldwide almost tripled, rising from 17 million to almost 50 million - [UNICEF](#). This group includes migrants, displaced persons, asylum seekers, and refugees. For these children, access to education in the host country is not only a pathway to future opportunities and integration but also a source of stability, psychosocial support, and protection.¹

However, when enrolling displaced children, schools face multiple challenges in assessing their skills and determining the best way to integrate and assign them to grades. First, displaced children could have experienced prolonged interruptions in schooling. Second, language differences often pose an immediate barrier. Third, displaced families may flee without or lose academic transcripts and school enrollment records. Finally, decisions to enroll and attend can suffer from high outside options arriving from families' vulnerability and the prevalence of child labor -[UNICEF](#).

These challenges highlight the importance of creating a well-developed system of reception of migrant students that properly assesses their educational background and enhances enrollment and attendance - see [Luciak \(2012\)](#) and [OECD. \(2019\)](#). In this context, placement exams can help assess students' pre-enrollment knowledge and guide integration into schools. However, depending on their design and implementation, they may also function as an unintended extra barrier to enrollment.

This paper examines how the introduction of a mandatory placement exam affects refugee children's school enrollment. We study the sudden and geographically concentrated displacement of Venezuelan migrants into Brazil triggered by Venezuela's economic and political crisis. In 2018 alone, more than 150,000 Venezuelans crossed into Brazil through the border state of Roraima. In Boa Vista, Roraima's capital and the main destination for migrants after crossing the border, the enrollment of Venezuelans jumped from virtually 0% in 2016 to almost 20% of the total student population in 2024. At the end of 2019, to standardize the enrollment process of foreign students without school documents, the local government implemented a mandatory placement exam for Venezuelan children aged 8 to 14. We employ a triple-difference framework that exploits both the policy's geographic implementation (exclusively in Boa Vista)

¹Galloway and Gjefsen ([2020](#)), Bridges and Walls ([2018](#)), and Felfe and Huber ([2015](#)).

and its targeted age range (exclusively 8 to 14).

Using national administrative enrollment data, we find that the policy has reduced school enrollment of refugee children by 30% between 2020 and 2024. Boys and girls were equally affected, and the effects were concentrated among ages 8 to 12, since 13- and 14-year-olds could enroll in state-managed elementary schools without the exam. Exploring administrative data on the exam registration list, we also found that the main barrier was not that refugee families failed to register their children for the exam, but rather the high exam absenteeism (30%). More specifically, the waiting time (on average, 30 days) and difficulty contacting families (some with only NGO phone numbers in their registry) increased the chances of exam absenteeism. Finally, Venezuelan children living in shelters, a specially vulnerable subgroup of the refugee population, were also more likely to miss the exam (by more than 20 percentage points).

Therefore, the results reveal that imposing placement exams before enrollment shifted the priority away from access to education. Therefore, prioritizing enrollment over the placement exam can help mitigate the policy’s negative effects. Importantly, improving exam capacity (i.e., reducing waiting times) and communication (e.g., publicly available exam schedule lists and direct administration of exams in refugee shelters) can reduce absenteeism and increase enrollment. Our results speak with other papers showing that removing barriers to exam access can substantially increase take-up, even when exams are high-stakes - Bulman (2015) (opening nearby SAT centers), Pallais (2015) (giving ACT takers more free score reports), and Muralidharan and Prakash (2017) (facilitating transportation to schools in India).

This paper also contributes to the yet limited literature on immigrants’ integration into host-country schools.² Conger (2013) finds that English language learners in Miami-Dade County who were assigned to lower grades—based on birthdate—experienced improved outcomes in mathematics but showed no gains in reading or retention. In contrast, Höckel and Schilling (2022) studies immigrant children in Hamburg, Germany, and finds that enrollment in preparatory language-learning classes led to lower test scores than direct placement in regular classrooms. Another set of papers studies how school tracking (vocational and academic tracks) and its timing af-

²See De Paola and Brunello (2016) for a review of the literature.

fect second-generation immigrants' performance and attainment - see Lüdemann and Schwerdt (2013), Jang and Brutt-Griffler (2023), and Cobb-Clark, Sinning, and Stillman (2012). These studies evaluate the effects of policies on migrants' educational outcomes after enrollment in developed countries. This paper examines a different, not studied policy and its impact on enrollment, a particularly relevant outcome in developing settings.

The rest of the paper is organized as follows. First, we describe the Venezuelan refugee inflow into Brazil, the public education system, and the policy change. The third section describes the data. Section 4 presents the empirical strategy, results, and mechanism analysis. Finally, Section 5 concludes.

2 Background

Venezuelan Refugee Inflow in Brazil

Venezuela's deep political and economic crisis led to a 65% decrease in its GDP between 2014 and 2019 and yearly inflation rates above 1000%.³ Human Rights Watch reported constant violations of human rights, including the persecution of journalists and civil society organizations and the capture of the judiciary by the government. UNHCR estimates that 7.7 million citizens emigrated, more than 84% to other countries in Latin America and the Caribbean.⁴

After 2017, with the worsening of the economic crisis in Venezuela, poorer immigrants began arriving at the border of Venezuela and Brazil in the state of Roraima.⁵ After crossing the border at the city of Pacaraima, immigrants go to Boa Vista, Roraima's capital and biggest city (400,000 people in 2020) - see Figure 1.

Between 2018 and 2019, almost 400,000 Venezuelan border crossings were registered in Roraima, representing more than 80% of the national inflow - see Figure 2. After a sharp drop in 2020 and 2021, when the border was closed due to the COVID-19 pandemic, entries surged again after 2023, demonstrating that the situation continues to evolve in the region.

³IMF statistics.

⁴See [R4V Platform](#) for statistics by hosting country.

⁵Source: [Ministry of Justice and Public Security report on Venezuelan Migration for April 2024](#).

Figure 1: Brazil-Venezuela Border and Roraima's Municipalities

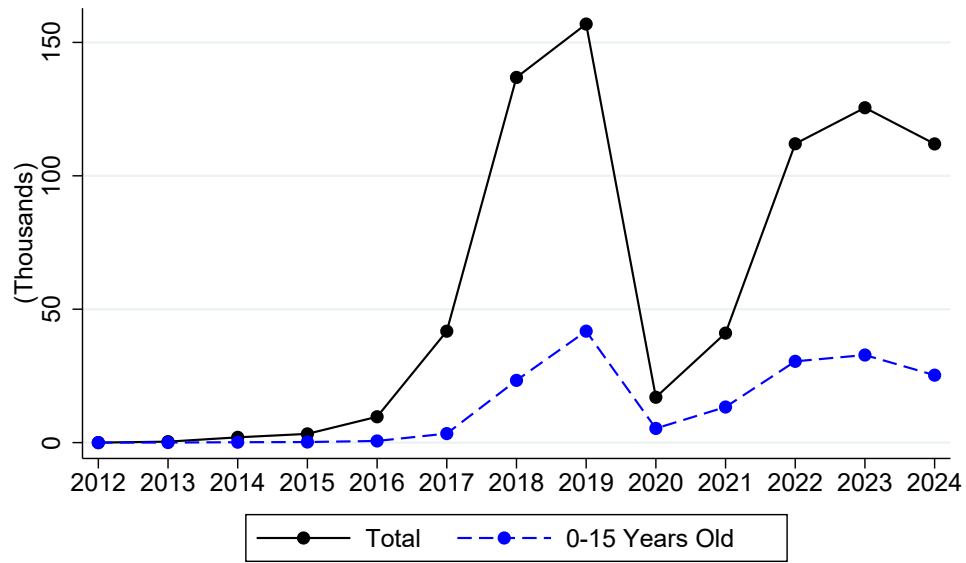


The federal government, along with local and state governments, NGOs, and the UNHCR, established shelters in Boa Vista to provide food and protection to Venezuelans. Refugees, disregarding their legal status, can access public schools.

Brazilian Public Education System

The Brazilian basic education cycle consists of 12 mandatory years of schooling (starting at age 6 up to 18). The school year in Brazil starts in February and ends in December. Municipality, State, and Federal Governments participate in the funding and management of public schools. Municipalities are responsible for early childhood education (children from 0 to 3 years old) and elementary education (6 to 14 years old). States also manage elementary schools, but are exclusively responsible for high schools (15 to 18 years old). Federal schools' enrollment represents a very small share of total enrollment.

Figure 2: Venezuelan Net Entrance Inflow in Roraima



Source: STI.

Figure 3: Schools and Refugee Shelters - Boa Vista (2019)

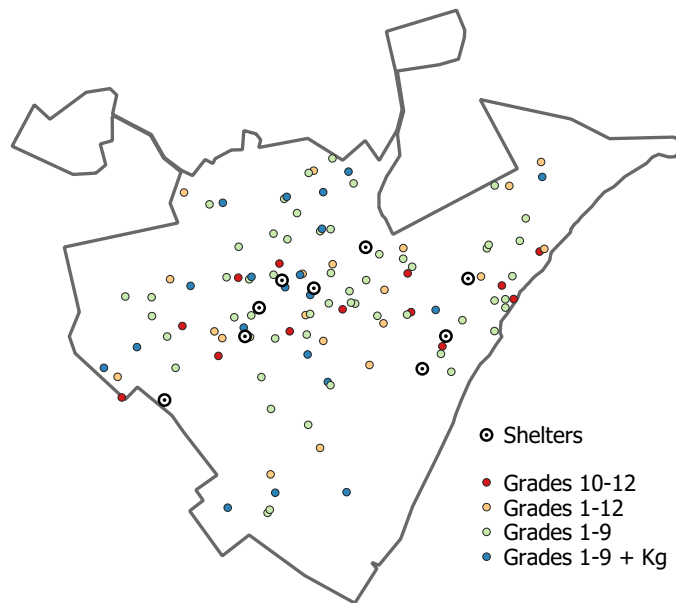


Figure 3 shows the geographic distribution of the different types of schools in the urban area of Boa Vista. In 2019, Boa Vista had 109 public schools (3 federal, 57 state, 49 municipal), 13 schools with only grades 10 to 12 (high schools), 20 with first to 12th grade, and 76 with students up to 9th grade.

To enroll a student, parents or responsible adults should go to any school or a special government office, bringing their ID, proof of residence with zip code, the child's ID,

and a telephone number for contact. The period of enrollment usually happens in January (it is still possible to enroll kids during the school year), and children who were already attending school are automatically enrolled for the following year. The enrollment process takes into account the number of spots available and where the child lives. The priority is to assign the student to the closest school with an available spot.

The Policy Change

Before September 2019, decisions regarding the grade placement of refugee children in Boa Vista's municipal schools were decentralized. Each school had discretion over whether and how to administer a placement exam, and these tests were often conducted in Portuguese.

In September 2019, the municipal education council enacted Resolution No. 01/2019, which standardized and centralized both enrollment and grade assignment procedures across all municipal schools. Under this new policy, placement exams became uniform, were offered in Spanish, and were administered by the municipal education secretariat staff in a designated office.

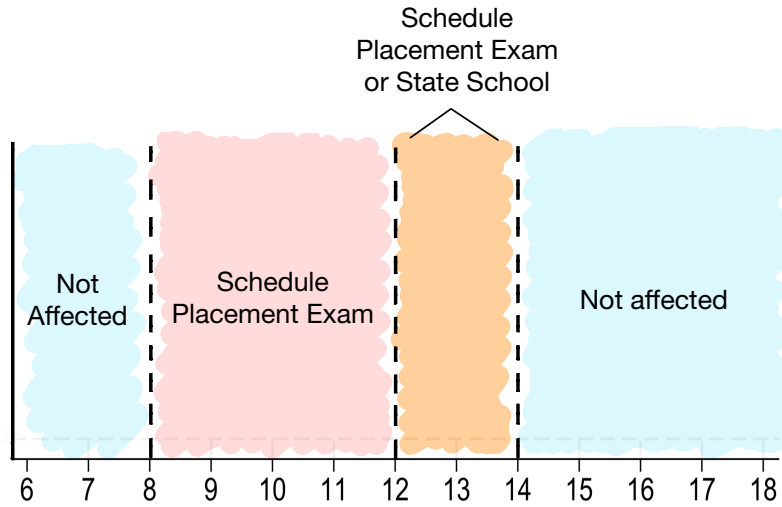
The exams followed a stepwise procedure: children were first tested on the grade level corresponding to their age—for instance, a 10-year-old took the fourth-grade exam. Those scoring below 50 percent in either language or mathematics would have to take the test corresponding to the grade below. The process was repeated until they passed or reached the first grade assignment. According to the resolution, all necessary exams were administered on the same day, and the entire procedure could last up to four hours.

The policy applied exclusively to Venezuelan children aged 8 to 14, as of March 31st of the enrollment year, who lacked school records or transcripts verifying their previous grade level. Importantly, it covered only municipal schools, which in this context serve grades 1 through 5. State-run schools, which offer grades six and above, were not subject to the placement exam policy, and their enrollment procedures remained decentralized and school-specific.

Consequently, the policy design generated three differently affected subgroups - see

Figure 4. First, children younger than eight or older than fourteen on the reference date were not required to take a placement exam. Second, those aged 12 to 14 were only partially affected, as they were already eligible, based on age, for enrollment in the 6th grade and could, therefore, bypass the municipal placement exam by enrolling directly in a state school. Finally, Venezuelan children aged 8 to 12 were fully subject to the policy, with no alternative non-placement exam schools to enroll in.

Figure 4: Placement Exam Policy Guideline by Age Group



Using the administrative data on the placement exam registration list from 2020 to 2025 (2021 not available), 99.45% of registered Venezuelan children are within the targeted ages (8 to 14 years old as of March 31st) - see Table 1. As expected, the number of 12 to 14-year-olds (partially affected) represents less than 14% of the total registry. The exam assignment based on age was also consistently implemented. Among the 8-year-olds, more than 98% took the 2nd grade exam, more than 97% of the 9-year-olds took the third grade, more than 99.5% of the 10-year-olds took the 4th grade exams, and more than 99.9% of the 11, 12, 13, and 14-year-olds took the 4th grade exam. Finally, no student was assigned to a grade that was not permitted according to the policy guideline.

Table 1: Number of Venezuelans in the Placement Exam Registry - per age

Age	Number of Venezuelans	%	Cum. %
6	2	0.04	0.04
7	14	0.25	0.28
8	1,223	21.54	21.82
9	1,287	22.66	44.48
10	1,394	24.55	69.03
11	1,010	17.78	86.81
12	450	7.92	94.73
13	188	3.31	98.05
14	101	1.78	99.82
15	9	0.16	99.98
16	1	0.02	100.00

Notes: data includes the years: 2020, 2022, 2023, 2024, 2025.

Source: administrative data from SMEC Boa Vista (RR).

3 Data

Administrative Public Schools' Enrollment

The School Census provides yearly school-level information about infrastructure, size, students, teachers, and staff for all schools in the country (the reference date is the end of May). Importantly, we observe students' race, gender, date of birth, and nationality. We measure the first year of enrollment as the first year students appear in the data.

Administrative Placement Exam Registration Data

Boa Vista's secretary of education keeps the list of all Venezuelan children who have scheduled a placement exam from 2020 to the beginning of 2025. It contains the registry date, the name of the parent and children, the date of birth, and the date, time, and place the exam was taken. For a subset of years, we also observe whether they live in a refugee shelter.

4 Empirical Strategy and Results

Effect on Enrollment

To evaluate whether the policy changed enrollment, we first compare the age distribution of enrolled students in Boa Vista before and after the policy implementation (September 2019). We grouped all enrollments from 2017 (when the inflow of school-age Venezuelans started) to 2019, and 2020 to 2024 (the last year available). According to Figure 5a, between 2017 and 2019, the enrollment share continuously decreased along with Venezuelans' age. After 2019, however, the curve showed a significant "mass" gap in the ages 8 to 12, which translated into higher shares for the other age groups.

Figure 5: Age PDF of Newly Enrolled Venezuelans - Boa Vista (RR)



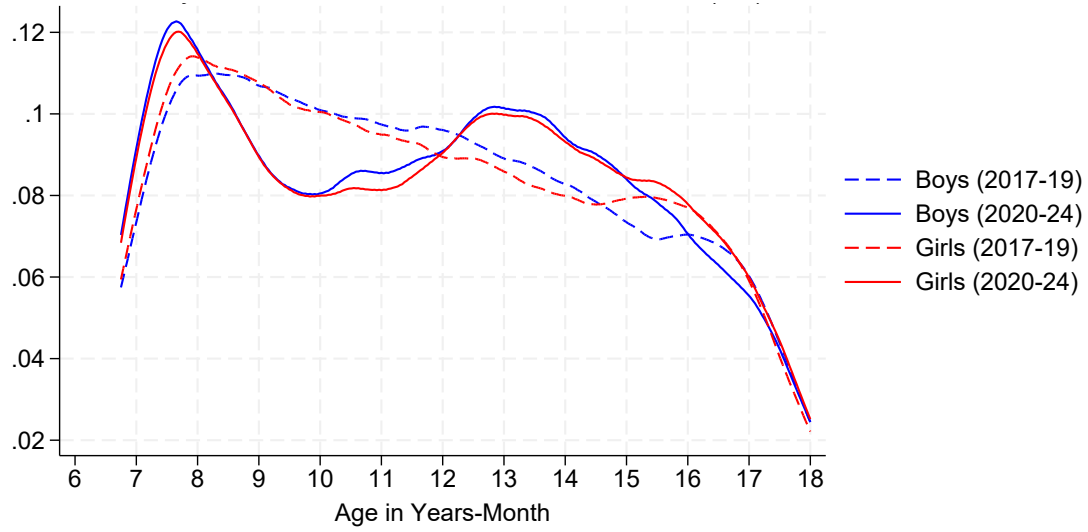
Notes: enrollment across all public schools. PDF estimated using an univariate kernel density estimation. *Source:* School Census (2017-2024).

Figure 5b shows that the change in the PDF shape started after the policy implementation at the end of 2019, and it is not driven by one specific year. Given that our data is a "photo" of enrollment on March 31st of each year, the 2020 data is a mixture of enrollment before and after the policy, and in fact, the 2020 PDF works as a transition between the 2019 and 2021. Therefore, it doesn't look like the change in the age PDF is due to some demographic change in the migration inflow that had already taken place before 2020. Moreover, it doesn't look like the pandemic years 2020 and 2021 are driving this difference, given that the 2022, 2023, and 2024 curves also present similar patterns.

When separating enrollment based on gender, we also observe that the change in

PDF shape also happened similarly for boys and girls - see Figure 6.

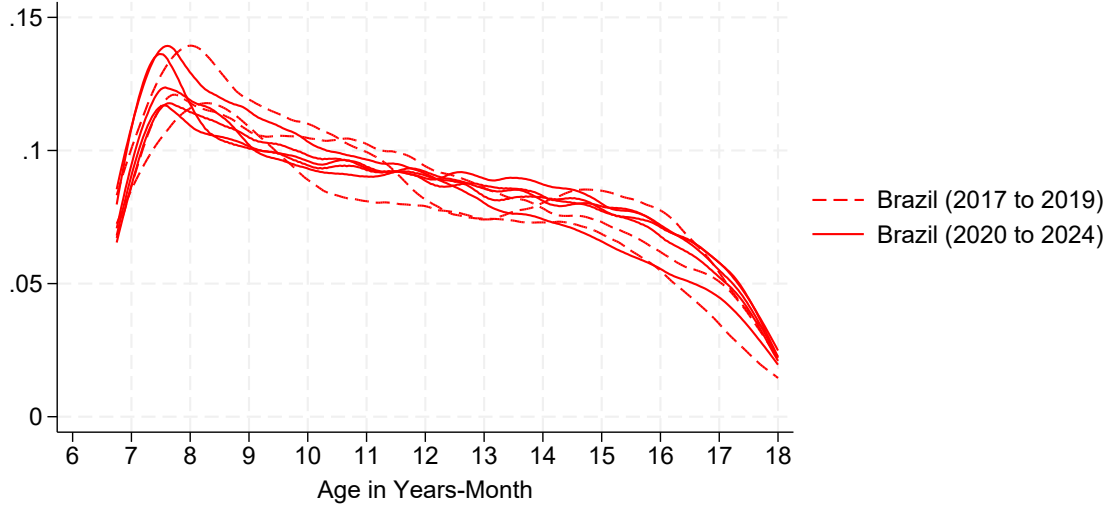
Figure 6: Age PDF of Newly Enrolled Venezuelans (Gender) - Boa Vista (RR)



Notes: enrollment across all public schools. PDF estimated using an univariate kernel density estimation. *Source:* School Census (2017-2024).

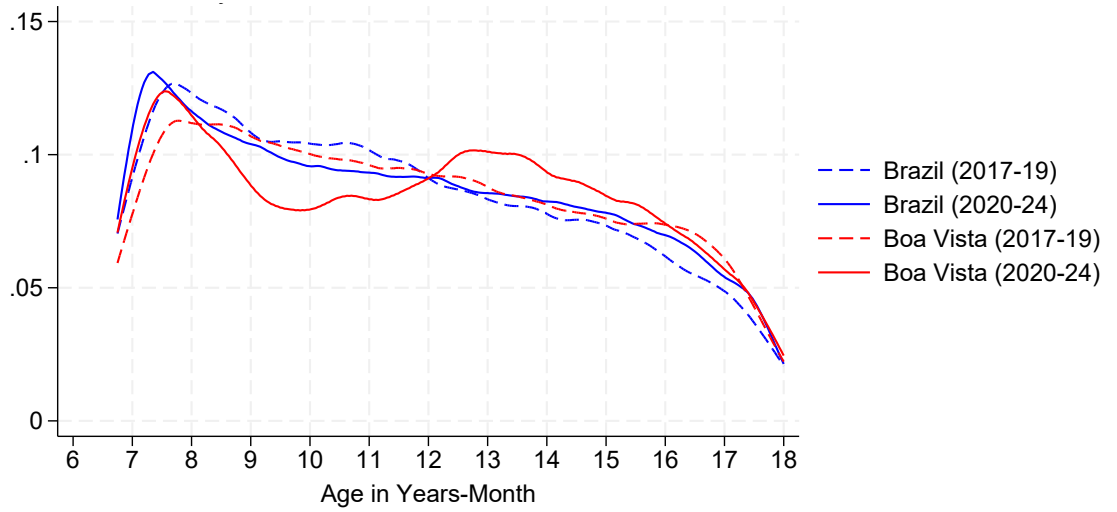
Another way to confirm the causal interpretation of the age density change is to compare it with the PDF of Venezuelan enrollment in the rest of the country, where no such placement exam requirement was introduced. According to Figures 7 and 8, the age distribution of Venezuelans enrolled in public schools in the rest of the country had a similar shape to that observed in Boa Vista before the policy. However, after 2019, only Boa Vista experienced such a change in enrollment age distribution. The similarity in the age distribution of enrolled Venezuelans between Boa Vista and the rest of the country is not surprising, given that Roraima accounted for more than 80% of Venezuelan entries into the country, and therefore the demographic composition of school-age Venezuelans entering the state is likely to be closer to the national one.

Figure 7: Age PDF of Newly Enrolled Venezuelans (Yearly) - Brazil (excluding RR)



Notes: enrollment across all public schools in Brazil excluding Roraima. PDF estimated using an univariate kernel density estimation. *Source:* School Census (2017-2024).

Figure 8: Age PDF of Newly Enrolled Venezuelans (Yearly) - Boa Vista Vs Brazil



Notes: enrollment across all public schools. "Brazil" category excludes Roraima state. PDF estimated using an univariate kernel density estimation. *Source:* School Census (2017-2024).

This setting allows us to estimate a Diff-in-Diff-in-Diff specification exploring the geographic concentration of the policy and its targeted age groups:

$$\begin{aligned}
 Y_{art} = & \alpha + \beta_1 \text{Post}_t + \beta_2 \text{BV}_r + \beta_3 \text{Group}_a + \\
 & \beta_4 (\text{Post}_t \times \text{BV}_r) + \beta_5 (\text{Post}_t \times \text{Group}_a) + \beta_6 (\text{BV}_r \times \text{Group}_a) + \\
 & \beta_7 (\text{Post}_t \times \text{BV}_r \times \text{Group}_a) + FE + \varepsilon_{art}
 \end{aligned} \tag{1}$$

Y_{art} is the number of Venezuelans enrolled in public schools in age group a (age in months) in region r (either Boa Vista or the rest of the country) in year t . $Post_t$ is a dummy after 2019. BV_r is a dummy for Boa Vista and $Group_a$ is a dummy for ages 8 to 14. The parameter of interest is β_7 . We explored different FE, including time, age-region, age-time, and region-time.

Table 2: Effect on Enrollment

	Y_{art}		$\ln(Y_{art})$	
	(1)	(2)	(3)	(4)
$Post_t \times BV_r \times Group_a$	-12.582*** (2.839)	-12.582*** (2.097)	-0.216*** (0.068)	-0.216*** (0.059)
$Post_t \times BV_r$	-68.065*** (2.301)		-1.666*** (0.049)	
$Post_t \times Group_a$	10.910*** (2.419)		0.076 (0.048)	
Observations	2,176	2,176	2,176	2,176
R^2	0.857	0.974	0.93	0.978
<i>Fixed-effects:</i>				
time	Y	N	Y	N
region-age	Y	Y	Y	Y
age-time	N	Y	N	Y
region-time	N	Y	N	Y

Notes: Average Y_{art} for Boa Vista: 21.6 (2017-24), 17.4 (2017-19), 24.1 (2020-24). Robust standard errors are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

According to Table 2, the policy reduced enrollment of Venezuelans in the target age group by approximately 22%. The results are robust to different fixed effects and outcome definitions. Moreover, to evaluate differential policy effects, we estimate equation 1 using the two samples, including the control ages and either the 8 to 11-year-olds or the 12 to 14-year-olds (who had the option to enroll in non-placement exam state-managed schools). As expected, the policy had no impact on the later group and decreased enrollment among highly affected age groups by 33% - see Table 3.

Table 3: Effect on Enrollment - By Age Group

	8-11 Years Old		12-14 Years Old	
	Y_{art}	$\ln(Y_{art})$	Y_{art}	$\ln(Y_{art})$
$\text{Post}_t \times \text{BV}_r \times \text{Group}_a$	-17.862*** (2.199)	-0.330*** (0.067)	-2.445 (2.42)	0.004 (0.076)
Observations	1,776	1,776	1,408	1,408
R^2	0.973	0.978	0.968	0.98
<i>Fixed-effects:</i>				
time	N	N	N	N
region-age	Y	Y	Y	Y
age-time	Y	Y	Y	Y
region-time	Y	Y	Y	Y

Notes: Robust standard errors are reported in parentheses.

*** p<0.01, ** p<0.05, * p<0.1.

Domestic migration of affected Venezuelans would inflate enrollment counts and lead us to overestimate the policy effects. Excluding of the state of all other municipalities of the state of Roraima from the "rest of Brazil" region can partially address this concern.⁶ However, to formally check this, we run the following Diff-Diff using data either just from Boa Vista (to confirm the policy effects) or from the rest of Brazil (to check for domestic migration):

$$Y_{at} = \alpha + \beta_1 \text{Post}_t + \beta_2 \text{Group}_a + \beta_3 (\text{Post}_t \times \text{Group}_a) + FE + \varepsilon_{at} \quad (2)$$

According to the results in Table 4, we confirm that the policy reduced enrollment in age group 8 to 11, and that we do not observe any increase in enrollment in this age group in other parts of the country where no such policy was implemented.

⁶In future analysis, we plan to use the other municipalities in Roraima, close to Boa Vista, to investigate domestic migration within the state.

Table 4: Diff-in-Diff Results

	Boa Vista		Rest of Brazil	
$\text{Post}_t \times \text{Group}_a$	-2.357 (1.47)		1.316 (2.143)	
$\text{Post}_t \times \text{Group (8-11)}_a$		-4.574*** (1.558)		1.723 (2.52)
$\text{Post}_t \times \text{Group (12-14)}_a$		1.901 (2.184)		0.534 (2.757)
Observations	1,088	1,088	1,088	1,088
R^2	0.835	0.837	0.977	0.977
<i>Fixed-effects:</i>				
time	N	N	N	N
region-age	Y	Y	Y	Y
age-time	Y	Y	Y	Y
region-time	Y	Y	Y	Y

Notes: Robust standard errors are reported in parentheses.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

What is the main barrier?

The placement exam consists of two parts: registering the children on the wait list and attending the exam. Therefore, the negative effect on enrollment can be due to families not registering children on the wait list or children failing to attend the exam. In this section, we discuss the main mechanisms/barriers behind the decrease in enrollment.

Between 2023 and 2024, around 3,300 children were registered on the placement exam wait list, and 30% (1,100) did not attend the exam. If we consider that all children would have been enrolled if they had attended, the negative effect of the policy would be erased for the entire period from 2020 to 2024.⁷ It is likely, consequently, that the registry includes families that would not enroll their children even in the absence of the policy, for example, transit families that planned to move to other parts of the country. Therefore, most of the challenges in refugee child enrollment are not related to families not registering kids (i.e., the registry not reaching enough families), but rather to the process after registration that is driving the high absenteeism.

⁷We only have data on exam attendance for 2023 and 2024

Next, we use the 2023 and 2024 administrative exams registry data to verify what explains the limited exam take-up by estimating equation 3. Our main explanatory variables measure the extent and quality of communication with the families to inform the dates of exams and how long the family is expected to wait for their children to take the exam.

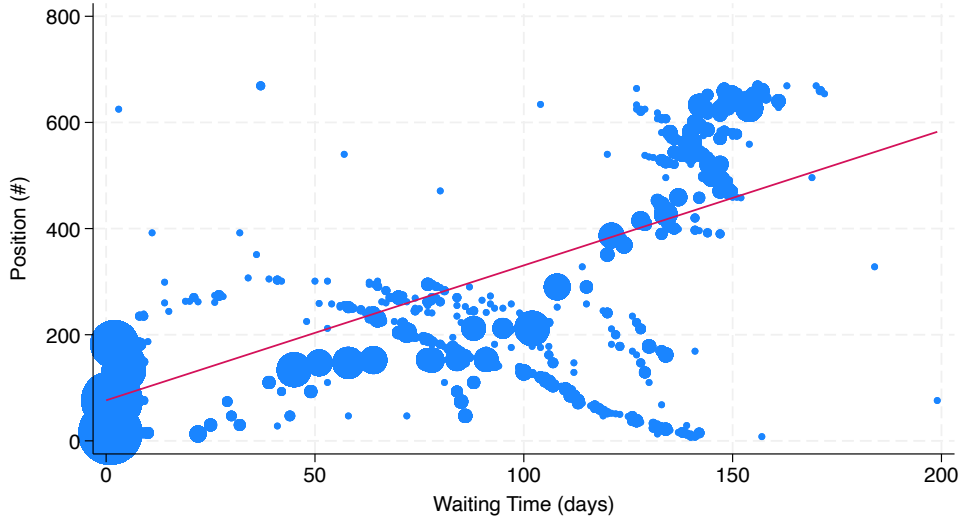
$$\begin{aligned} \text{Missed Exam}_i = & \alpha + \beta_1 \text{Wait List Position}_i \\ & + \beta_2 \text{Phone}_i + \beta_3 \text{Multiple Phones}_i + \beta_4 \text{NGO Phone}_i \\ & + FE + \varepsilon_i. \end{aligned} \quad (3)$$

The dependent variable is *Missed Exam_i*, a dummy for student "i" not taking the exam. *Phone_i* is a dummy equal to one if the individual provided at least one valid phone number at registration, and *Multiple Phones_i* indicates whether more than one phone number was annotated. *NGO Phone_i* is a dummy equal to one if one of the contact numbers belongs to a humanitarian organization. Around 8% of registered families rely on NGO phone numbers that can operate as an intermediary agent in the communication.

The time waited between registering the children and taking the exam is correlated with how easy it is to contact the family (exams can be postponed until communication is complete). Therefore, to measure how the capacity/speed of the exam scheduling process affects absenteeism, we instead use the variable *Wait List Position_i*, which measures how far down the queue the individual was when registered for the exam. Higher positions reflect longer waiting times - see Figure 9. This proxy for waiting time is less likely to be correlated with other family and children characteristics. Finally, we explore different sets of fixed effects (*FE*), including year, age, and month-year of registry. Our preferred specification compares students with the same age and registered in the same month and year.

According to the results in Table 5, the higher the position in the waiting list, the higher the chances of absenteeism. In addition, as expected, the existence of a phone number in the registry and the registry of multiple phone numbers can facilitate communication and are associated with smaller absenteeism. However, having an NGO phone number is associated with higher chances of absenteeism, likely because this

Figure 9: Position in the Wait List Vs Waiting Time



Note: red curve represents the fit/regression line between the two variables and a constant. *Source:* Placement Exam Registry for 2023 and 2024.

variable captures other socio-economic variables of student "i" and their family. This becomes clear once we add a dummy for whether children "i" live in a shelter (only available for a subsample), and the coefficient associated with NGO phone number decreases and loses significance - see column (4) of Table 5.⁸

Absenteeism is considerably higher (around 20 percentage points) for refugee children living in shelters. The addition of this variable, however, does not change the effects of wait list position and having multiple contact phone numbers, reinforcing the importance of those two logistical components in mediating exam attendance. Finally, the interaction of phone communication variables and position in the wait list (results not reported in this draft) is not significant, and therefore, indicates that both sets of variables matter by themselves.

⁸The results are similar when estimating equation 3 using logit - see Table 6 in the Appendix.

Table 5: What Explains Exam Absenteeism

Dependent Variable:	Missed Exam (Dummy)			
	(1)	(2)	(3)	(4)
Wait List Position	0.0006*** [0.13***] (0.0000)	0.0006*** [0.13***] (0.0000)	0.0006*** [0.13***] (0.0001)	0.0009*** [0.29***] (0.0002)
Have Phone Number	-0.1245*** (0.0464)	-0.1268*** (0.0462)	-0.1059** (0.0450)	-0.2017 (0.1352)
Have 2+ Phone Numbers	-0.0958*** (0.0194)	-0.0913*** (0.0194)	-0.0609*** (0.0189)	-0.0702** (0.0318)
Have NGO Phone Number	0.1680*** (0.0246)	0.1717*** (0.0245)	0.1746*** (0.0238)	0.0549 (0.0375)
Live in a Shelter				0.2055*** (0.0300)
Observations	3,298	3,298	3,296	1,472
R-squared	0.3407	0.3479	0.4006	0.3078
<i>Fixed Effects:</i>				
Year FE	Y	Y	N	N
Age FE	N	Y	Y	Y
Month-Year Registered FE	N	N	Y	Y
Avg. Missed Exam	0.299	0.299	0.299	0.496
Avg. Wait List Position	222.2	222.2	222.3	325.3
Avg. Have Phone Number	0.980	0.980	0.980	0.993
Avg. Have 2+ Phones Numbers	0.130	0.130	0.130	0.147
Avg. Have NGO Phone Number	0.0797	0.0797	0.0798	0.147
Avg. Live in shelter				0.357

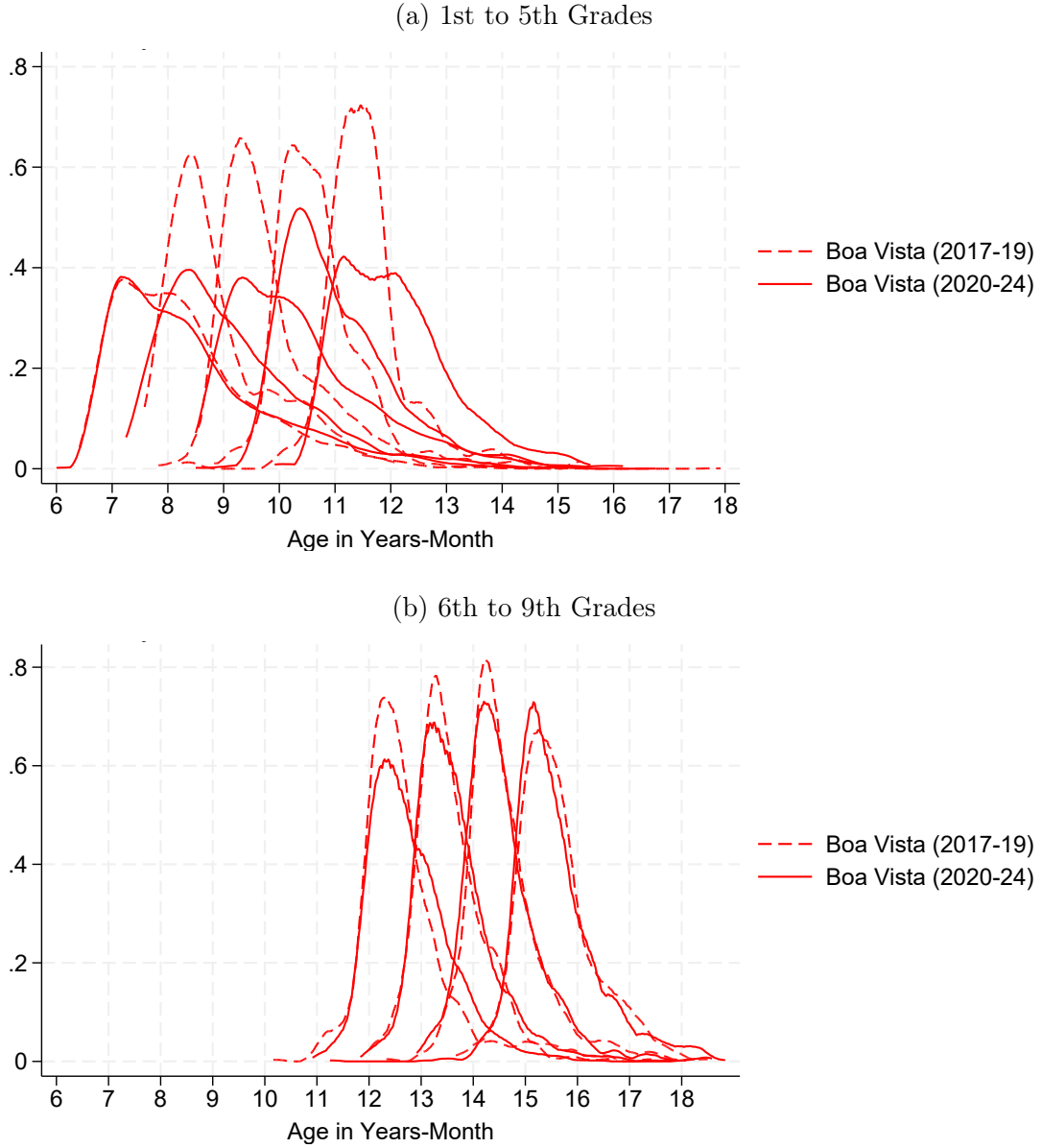
Notes: β multiplied by the average *Wait List Position* in brackets. Standard errors clustered at the country level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Grade Assignment

Finally, we examine how grade assignment changed after the policy. More specifically, the policy could also have changed the grades that Venezuelans are assigned to, potentially affecting age distortion rates (share of students at least two years older than expected for that grade) among newly enrolled migrants.

Looking at the age distribution of Venezuelans enrolled in each grade, we can also observe that age distortion rates increased after the policy implementation, especially for grades 2 to 5. In other words, the pull of students being enrolled at each grade from 2nd to 5th got on average older, due to the expansion in the left tail of the age distribution. As expected, given that the policy had a limited effect on 12-year-olds or older migrants, we don't observe relevant changes for grades 6th to 9th.

Figure 10: Age Density Per Grade - Boa Vista (RR)



Notes: enrollment across all public schools. PDF estimated using an univariate kernel density estimation. *Source:* School Census (2017-2024).

At the same time that the policy changed the exam language from portuguese to spanish, the negative enrollment effect likely came with a change in the selection of Venezuelan students. Moreover, given that before the policy grade assignment decisions were purely school-specific, it is not clear whether previous assignments were more generous or more strict. However, it is still relevant that the policy likely affected age-distortion.

5 Conclusion

Between 2010 and 2024, the number of displaced children worldwide rose from 17 to nearly 50 million. Access to education offers these children stability, protection, and a path to integration, yet schools face challenges such as interrupted schooling, language barriers, and missing documents. Placement exams can assist schools in assessing migrants' pre-enrollment skills and guide grade assignment. However, depending on their design and implementation, they may also function as an unintended barrier, reducing school enrollment.

This paper studies how a mandatory placement exam affected refugee children's school enrollment in Boa Vista, Brazil, following the large inflow of Venezuelan migrants after 2017. The local government introduced a mandatory placement exam at the end of 2019 for all Venezuelan children aged 8–14 without school. Using a triple-difference framework, we exploit both the policy's geographic restriction to Boa Vista and its targeted age group to estimate its impact on school enrollment using national administrative data.

We find that the introduction of the exam in Boa Vista reduced school enrollment among refugee children in the targeted age group. We estimate that the policy affected the enrollment of around 1,000 Venezuelan children between 2020 and 2024. Exploring unique administrative data on the exams' registration list, we also found that the main barrier was not that refugee families failed to register their children for the exam, but rather the waiting time (on average, 30 days) and difficulty in contacting families, which increased the exam absenteeism. Finally, Venezuelan children living in shelters, a specially vulnerable subgroup of the refugee population, were also more likely to miss the exam (by more than 20 percentage points).

This project is among the first economic papers studying the integration of refugee children into the host country's public education. Together, the results reveal that local government should consider changes in the way placement exams are applied. Importantly, improving exam capacity (i.e., reducing waiting times) and communication (such as using alternative strategies, such as publicly available exam schedule lists) can reduce absenteeism and increase enrollment. Conducting placement exams in shelters to reach out-of-school children could also improve enrollment, especially

among this vulnerable subgroup of refugees most adversely affected by the policy.

Finally, this paper did not evaluate the effect of grade assignment. More specifically, whether downgrading/age distortion, is beneficial for refugees education outcomes (such as test scores, attendance and retention). Future work in understanding refugee students integration after enrollment is needed.

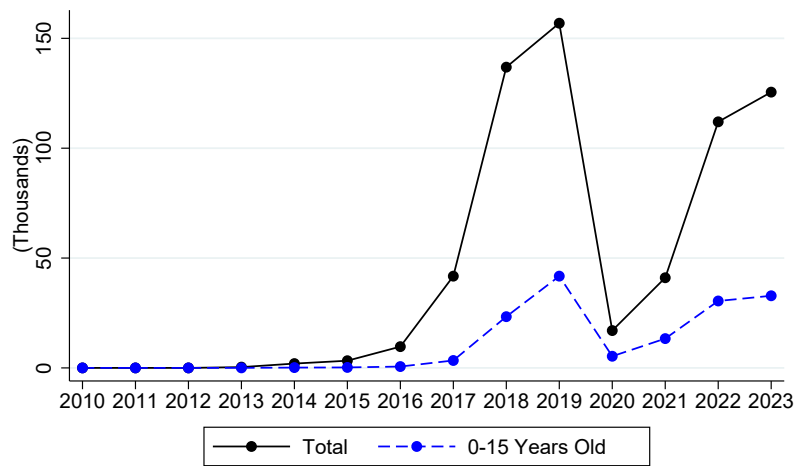
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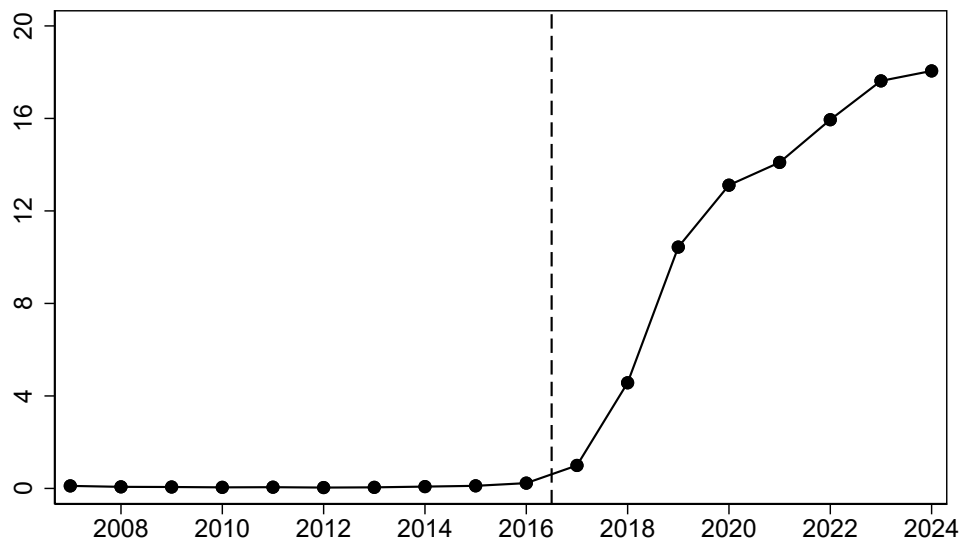
Appendix

Figure 11: Venezuelan Net Entrance Flow - Roraima's Border



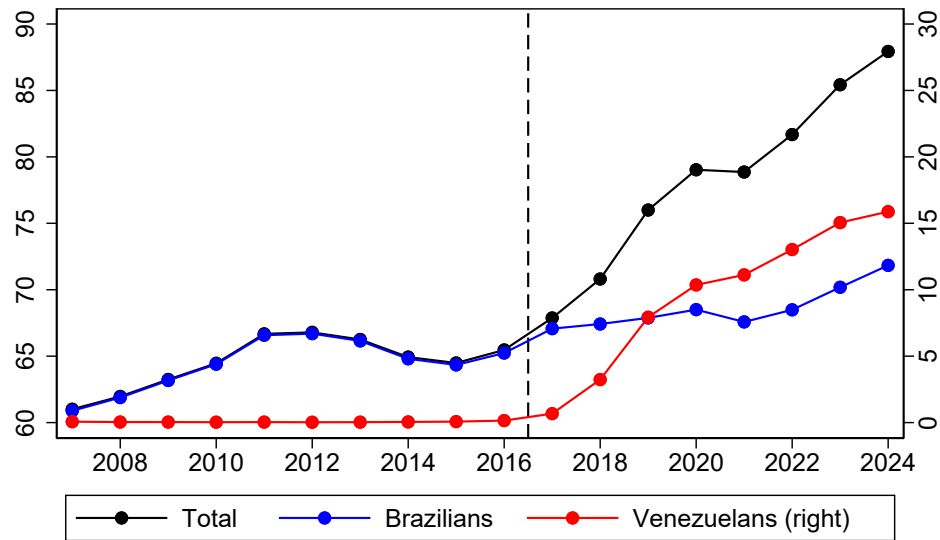
Source: STI.

Figure 12: % Venezuelan Students in Public Schools - Boa Vista (RR)



Source: School Census (2017-2024).

Figure 13: Number (1,000) of Venezuelan Students in Public Schools - Boa Vista (RR)



Note: Includes all public schools from grades 1 to 12, kindergarten to Nursery schools. *Source:* School Census (2017-2024).

Figure 14: Age Histogram of Newly Enrolled Venezuelans - Boa Vista (RR)



Notes: enrollment across all public schools. PDF estimated using an univariate kernel density estimation. *Source:* School Census (2017-2024).

Table 6: What Explains Exam Absenteeism - Logit

Dependent Variable:	Missed Exam (Dummy)			
	(1)	(2)	(3)	(4)
Wait List Position	0.0031*** (0.0003)	0.0031*** (0.0003)	0.0038*** (0.0007)	0.0044*** (0.0009)
Have Phone Number	-0.8201** (0.3440)	-0.8453** (0.3505)	-0.8061** (0.3432)	-1.3074 (0.9028)
Have 2+ Phone Numbers	-0.5880*** (0.1393)	-0.5723*** (0.1409)	-0.5194*** (0.1535)	-0.4213** (0.1841)
Have NGO Phone Number	0.9775*** (0.1630)	1.0179*** (0.1619)	1.1774*** (0.1762)	0.2808 (0.2364)
Live in a Shelter				1.1905*** (0.1852)
Observations	3,298	3,298	2,785	1,375
<i>Fixed Effects:</i>				
Year FE	Y	Y	N	N
Age FE	N	Y	Y	Y
Month-Year Registered FE	N	N	Y	Y
Avg. Missed Exam	0.299	0.299	0.299	0.496
Avg. Wait List Position	222.2	222.2	222.3	325.3
Avg. Have Phone Number	0.980	0.980	0.980	0.993
Avg. Have 2+ Phones Numbers	0.130	0.130	0.130	0.147
Avg. Have NGO Phone Number	0.0797	0.0797	0.0798	0.147
Avg. Live in shelter				0.357

Notes: β multiplied by the average *Wait List Position* in brackets. Standard errors clustered at the country level in parentheses. *** p<0.01, ** p<0.05, * p<0.1